

No Demonstrated Effect of Pre-enucleation Irradiation on Survival of Patients With Uveal Melanoma

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- **PURPOSE:** To evaluate the hypothetical effect of pre-enucleation irradiation on survival of patients with uveal melanoma.
- **METHODS:** In a prospective study between 1978 and 1990, 145 patients with uveal melanoma were treated by irradiation in two fractions of 4 Gy before enucleation. A historical control group of 89 patients with uveal melanoma treated by enucleation alone was operated on between 1971 and 1990. Patients were followed up until December 1992 or until death. The mean follow-up period was 65 months in the irradiated group and 88 months in the control group.
- **RESULTS:** The preoperatively irradiated group of patients showed no significant improvement of the survival rate after 7½ years (75.9%) compared with the control group (72.1%). Preoperative irradiation was not associated with survival ($P = .93$), as assessed by Cox proportional hazard analysis, adjusted for age, gender, tumor location, tumor size, cell type, and year of enucleation.

Women in both the irradiated and control groups had a better prognosis than men ($P = .002$).

- **CONCLUSION:** Preoperative irradiation in this nonrandomized study had no effect on survival of patients with uveal melanoma.

WHETHER MICROMETASTASIS CAN BE INDUCED by physical manipulation of an eye with uveal melanoma during the enucleation procedure, resulting in increased risk for tumor-related death, remains to be determined.^{1,2} In animal models, melanoma cells can get into the circulation by manipulation of the tumor-containing eye.³ To reduce this hypothetical effect of the surgical procedure on the survival of uveal melanoma patients, preoperative radiotherapy has been introduced.^{2,4-7} Irradiation of the tumor before surgery might alter the tumor cells, thus preventing the implantation and growth of the circulating tumor cells after the enucleation procedure.⁸⁻¹¹ In the reported clinical studies no beneficial effect of pre-enucleation radiotherapy has been demonstrated.^{4,12-14} Char and associates⁴ studied a small series of nonrandomly selected patients and found that preoperative irradiation with 20 Gy had a worse prognosis compared with patients treated by enucleation alone.

The purpose of our study was to evaluate the long-term effect of low-dose pre-enucleation irradiation on the survival of a large nonselected series of patients with uveal melanoma compared with a control group of patients treated by enucleation alone. To date this study is the largest series of nonselected

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